EXPERIMENT:9

AIM: To write a C program to decrypt a message encrypted using the Playfair Cipher.

CODE:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int main() {
   char s[] =
"KXJEYUREBEZWEHEWRYTUHEYFSKREHEGOYFIWTTTUOLKSYCAJPOBOTEIZON
TXBYBNTGONEYCUZWRGDSONSXBOUYWRHEBAAHYUSEDQ";
   char key[6][6] = {
      {'K','E','Y','W','O'},
      {'R','D','A','B','C'},
      {'F','G','H','I','L'},
      {'M','N','P','Q','S'},
      {'T','U','V','X','Z'}
   };
   int i, j, r1, c1, r2, c2, x, y;

   for(i = 0; s[i] && s[i+1]; i += 2) {
      r1 = c1 = r2 = c2 = -1;
      for(x = 0; x < 5; x++)
         for(y = 0; y < 5; y++) {
            if(key[x][y] == s[i]) r1=x, c1=y;
            if(key[x][y] == s[i+1]) r2=x, c2=y;
         }
```

```
        if(r1 == r2)
            printf("%c%c", key[r1][(c1+4)%5], key[r2][(c2+4)%5]);
        else if(c1 == c2)
            printf("%c%c", key[(r1+4)%5][c1], key[(r2+4)%5][c2]);
        else
            printf("%c%c", key[r1][c2], key[r2][c1]);
    }
    return 0;
}
```

OUTPUT:

